COMSATS UNIVERSITY ISLAMABAD

ATTOCK CAMPUS

Computer Science Department

Course Title: Information Security

Assighment 1

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1. Caesar Cipher:

The **Caesar Cipher** is one of the earliest and simplest encryption techniques in cryptography. It is a type of substitution cipher in which each letter in the plaintext is shifted by a fixed number of positions in the alphabet.

2. Code:

```python
# Function to encrypt text using Caesar Cipher
def caesar_encrypt(text, shift):  1 usage
    encrypted_text = ""

    for char in text:
        # Check if character is uppercase letter
        if char.isupper():
            # Convert letter to ASCII, shift it, wrap around using modulo
            encrypted_text += chr((ord(char) - 65 + shift) % 26 + 65)

        # Check if character is lowercase letter
        elif char.islower():
            encrypted_text += chr((ord(char) - 97 + shift) % 26 + 97)

        # If space or special character, keep it unchanged
        else:
            encrypted_text += char

    return encrypted_text


# Function to decrypt text using Caesar Cipher
def caesar_decrypt(ciphertext, shift):  1 usage
    decrypted_text = ""

    for char in ciphertext:
        if char.isupper():
            decrypted_text += chr((ord(char) - 65 - shift) % 26 + 65)

        elif char.islower():
```

```python
        elif char.islower():
            decrypted_text += chr((ord(char) - 97 - shift) % 26 + 97)

        else:
            decrypted_text += char

    return decrypted_text


# Main Program
if __name__ == "__main__":
    message = input("Enter your message: ")
    shift = int(input("Enter shift value: "))

    encrypted = caesar_encrypt(message, shift)
    print("Encrypted Message:", encrypted)

    decrypted = caesar_decrypt(encrypted, shift)
    print("Decrypted Message:", decrypted)
```

3. Line-by-Line Explanation

➢ Encryption Function

- **def caesar_encrypt(text, shift):**
  Defines a function that takes a message and shift value.
- **encrypted_text = ""**
  Creates an empty string to store encrypted output.
- **for char in text:**
  Loops through each character in the input message.
- **if char.isupper():**
  Checks if the character is uppercase.
- **chr((ord(char) - 65 + shift) % 26 + 65)**
  Converts character to ASCII using ord(), shifts it, uses modulo 26 to wrap around, and converts back using chr().
- **elif char.islower():**
  Same logic for lowercase letters using ASCII value 97.
- **else:**
  Keeps spaces and special characters unchanged.

➢ Decryption Function

- **def caesar_decrypt(ciphertext, shift):**
  Defines decryption function.
- It subtracts the shift instead of adding it.
- % 26 ensures wrap-around within the alphabet.

➢ Main Block

- Takes user input
- Calls encryption function
- Prints encrypted message
- Calls decryption function
- Prints original message

4. Output:

```
Enter your message: my name is noor
Enter shift value: 4
Encrypted Message: qc reqi mw rssv
Decrypted Message: my name is noor

Process finished with exit code 0
```